Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.001 with the Adam optimizer. All experiments were run on a single GPU with 16 GB of memory.

2 Results

Figure 1 shows the test accuracy during training.

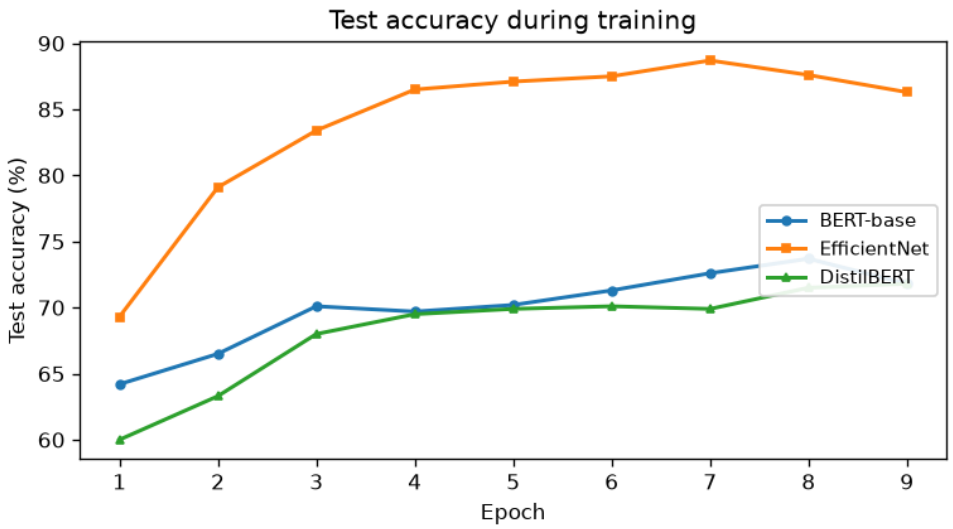


Figure 1: Test accuracy during training for the models in this experiment.

At epoch 9, EfficientNet is 30.3 points better than DistilBERT. EfficientNet outperforms BERT-base by about 5 percentage points at epoch 1 (Figure 1). BERT-base attains a test accuracy of about 70% at epoch 7 (Figure 1). EfficientNet outperforms BERT-base by 3.5 percentage points at epoch 1.

Figure 2 shows the test accuracy during training.

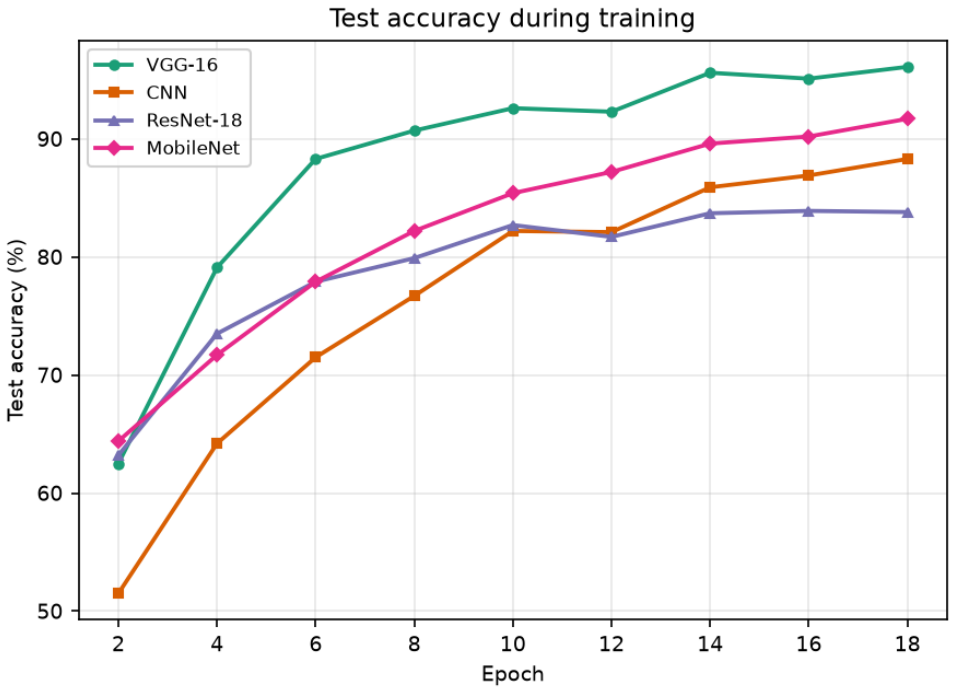


Figure 2: Test accuracy during training for the models in this experiment.

As shown in Figure 2, VGG-16 outperforms CNN by 32.7 percentage points at epoch 16. At epoch 12, VGG-16 achieves 116.8% test accuracy (Figure 2).

Figure 3 shows the test accuracy by dataset.

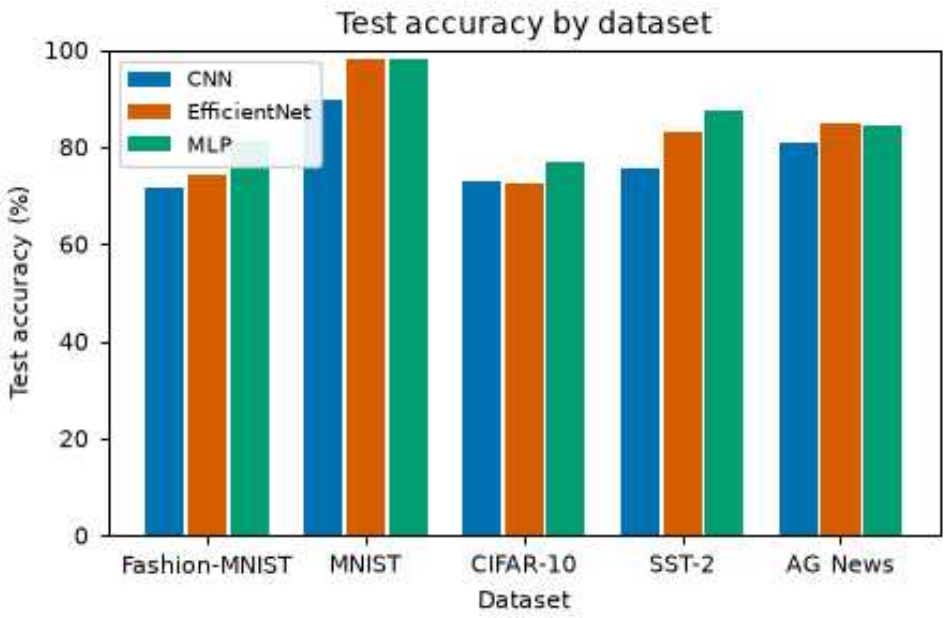


Figure 3: Test accuracy by dataset for the models in this experiment.

CNN reaches 69.6% test accuracy on MNIST. EfficientNet achieves 48.2% test accuracy on MNIST. As shown in Figure 3, EfficientNet achieves 72.1% test accuracy on Fashion-MNIST. CNN reaches 92.7% test accuracy on CIFAR-10.

3 Discussion

Overall, the results are in line with what we expected from the literature.